

LAB 7

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CLASS: BSE-5A
SUBJECT: CC

LAB TITLE: Environment Variables, PATH, UFW, and SSH Key Authentication

TASK:

Task 1: Print & Filter Environment Variables

Goal:

Show environment variables and filter them using grep.

1. Print all environment variables

```
hajra@ubuntu-lab:~$ printenv
SHELL=/bin/bash
PWD=/home/hajra
LOGNAME=hajra
XDG_SESSION_TYPE=ttty
HOME=/home/hajra
LANG=en_GB.UTF-8
LS_COLORS=rs=0:di=01;34;ln=01;36;mh=00;pi=40;33;so=01;35;do=01;35;bd=40;33;01;cd=40;33;01;or=40;31;01;mi=00;su=37;41;sg=30;43;ca=00;tw=30;42;ow=34;42;st=37;44;ex=01;32;*.tar=01;31;*.tgz=01;31;*.arc=01;31;*.arj=01;31;*.taz=01;31;*.lha=01;31;*.lzh=01;31;*.lzm=01;31;*.tlz=01;31;*.txz=01;31;*.tzo=01;31;*.t7z=01;31;*.zip=01;31;*.z=01;31;*.dz=01;31;*.gz=01;31;*.lrz=01;31;*.lzh=01;31;*.lzo=01;31;*.xz=01;31;*.zst=01;31;*.tzt=01;31;*.bz2=01;31;*.bz=01;31;*.tbz=01;31;*.tbz2=01;31;*.tz=01;31;*.deb=01;31;*.rpm=01;31;*.jar=01;31;*.war=01;31;*.ear=01;31;*.sar=01;31;*.rar=01;31;*.alz=01;31;*.ace=01;31;*.zoo=01;31;*.cpio=01;31;*.7z=01;31;*.rz=01;31;*.cab=01;31;*.wim=01;31;*.swm=01;31;*.dwm=01;31;*.esd=01;31;*.avif=01;35;*.jpg=01;35;*.jpeg=01;35;*.mjpg=01;35;*.mjpeg=01;35;*.gif=01;35;*.bmp=01;35;*.pbm=01;35;*.pgm=01;35;*.ppm=01;35;*.tga=01;35;*.xbm=01;35;*.xpm=01;35;*.tif=01;35;*.tiff=01;35;*.png=01;35;*.svg=01;35;*.svgz=01;35;*.ang=01;35;*.pex=01;35;*.mov=01;35;*.mpg=01;35;*.mpeg=01;35;*.m2v=01;35;*.mkv=01;35;*.webm=01;35;*.webp=01;35;*.ogm=01;35;*.mp4=01;35;*.m4v=01;35;*.mpdv=01;35;*.vob=01;35;*.qt=01;35;*.nuv=01;35;*.wmv=01;35;*.asf=01;35;*.rm=01;35;*.rmvb=01;35;*.flc=01;35;*.avi=01;35;*.fli=01;35;*.flv=01;35;*.gl=01;35;*.dl=01;35;*.xcf=01;35;*.xwd=01;35;*.yuv=01;35;*.cgm=01;35;*.emf=01;35;*.ogv=01;35;*.ogx=01;35;*.aac=00;36;*.au=00;36;*.flac=00;36;*.m4a=00;36;*.mid=00;36;*.midi=00;36;*.mka=00;36;*.mp3=00;36;*.mpc=00;36;*.ogg=00;36;*.ra=00;36;*.wav=00;36;*.oga=00;36;*.opus=00;36;*.spx=00;36;*.xspf=00;36;*.~00;90;*.#00;90;*.bak=00;90;*.crdownload=00;90;*.dpkg-dist=00;90;*.dpkg-new=00;90;*.dpkg-old=00;90;*.dpkg-tmp=00;90;*.old=00;90;*.orig=00;90;*.part=00;90;*.rej=00;90;*.rpmnew=00;90;*.rpmorig=00;90;*.rpmsave=00;90;*.swp=00;90;*.tmp=00;90;*.ucf-dist=00;90;*.ucf-new=00;90;*.ucf-old=00;90;
SSH_CONNECTION=192.168.161.1 50467 192.168.161.129 22
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=hajra
SHELL=/bin/bash
XDG_SESSION_ID=3
XDG_RUNTIME_DIR=/run/user/1000
SSH_CLIENT=192.168.161.1 50467 22
XDG_DATA_DIRS=/usr/share/gnome:/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
dbus_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
SSH_TTY=/dev/pts/0
=/usr/bin/printenv
```

2. Filter for SHELL, HOME, USER

```
hajra@ubuntu-lab:~$ printenv | grep SHELL
SHELL=/bin/bash
hajra@ubuntu-lab:~$ printenv | grep HOME
HOME=/home/hajra
hajra@ubuntu-lab:~$ printenv | grep USER
USER=hajra
```

Task 2: Export DB_ Variables Temporarily and Observe Scope

Goal:

Create environment variables with export in the current shell, verify them, then close the shell and confirm the variables disappear.

1. Define all DB_ variables
2. Echo the three variables
3. Show all DB_ variables using grep
4. Close and reopen the shell — verify variables are gone

Screenshot:

```
hajra@ubuntu-lab:~$ export DB_URL="postgres://db.example.local:5432/mydb"
hajra@ubuntu-lab:~$ export DB_USER="labuser"
hajra@ubuntu-lab:~$ export DB_PASSWORD="labpass123"
hajra@ubuntu-lab:~$
hajra@ubuntu-lab:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
hajra@ubuntu-lab:~$ echo "$DB_USER"
labuser
hajra@ubuntu-lab:~$ echo "$DB_PASSWORD"
labpass123
hajra@ubuntu-lab:~$
hajra@ubuntu-lab:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
hajra@ubuntu-lab:~$
hajra@ubuntu-lab:~$ exit
logout
Connection to 192.168.161.129 closed.
PS C:\Users\VP> ssh hajra@192.168.161.129
hajra@192.168.161.129's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat 15 Nov 07:58:52 UTC 2025

System load:  0.83          Processes:      286
Usage of /:   47.1% of 37.29GB Users logged in:   1
Memory usage: 55%          IPv4 address for ens33: 192.168.161.129
Swap usage:   16%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Sat Nov 15 07:51:15 2025 from 192.168.161.1
hajra@ubuntu-lab:~$ echo "$DB_URL"
hajra@ubuntu-lab:~$
hajra@ubuntu-lab:~$ printenv | grep '^DB_'
hajra@ubuntu-lab:~$
```

Task 3: Make DB_ Variables Persistent in ~/.bashrc*

Goal:

Add DB_* variables to ~/.bashrc, reload the file, and verify persistence.

1. Add the DB_ variables inside ~/.bashrc*
2. Source ~/.bashrc and verify
3. Close and reopen the terminal — verify persistence

Screenshot:

```
hajra@ubuntu-lab:~$ vim ~/.bashrc
hajra@ubuntu-lab:~$
hajra@ubuntu-lab:~$ source ~/.bashrc
hajra@ubuntu-lab:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
hajra@ubuntu-lab:~$ echo "$DB_USER"
labuser
hajra@ubuntu-lab:~$ echo "$DB_PASSWORD"
labpass123
hajra@ubuntu-lab:~$
hajra@ubuntu-lab:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
hajra@ubuntu-lab:~$ exit
logout
Connection to 192.168.161.129 closed.
PS C:\Users\VP> ssh hajra@192.168.161.129
hajra@192.168.161.129's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat 15 Nov 08:51:03 UTC 2025

System load:  0.67          Processes:      286
Usage of /:   47.1% of 37.29GB Users logged in:   1
Memory usage: 62%          IPv4 address for ens33: 192.168.161.129
Swap usage:   13%

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The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Sat Nov 15 07:58:53 2025 from 192.168.161.1
hajra@ubuntu-lab:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
hajra@ubuntu-lab:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
hajra@ubuntu-lab:~$
```

Task 4: System-wide Environment Variable, Welcome Script, and PATH

Goal:

Add a system-wide Class variable to `/etc/environment`, view `PATH`, create a welcome script, make it executable, and add a `PATH` entry to `~/.bashrc` so the script can run without `./`.

1. View `/etc/environment`

```
hajra@ubuntu-lab:~$ sudo cat /etc/environment
[sudo] password for hajra:
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
```

2. Show current `PATH`

```
hajra@ubuntu-lab:~$ echo "$PATH"
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
```

3. Edit `/etc/environment` to add Class variable

```
hajra@ubuntu-lab:~$ sudo cat /etc/environment
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
Class="CC-5A"
hajra@ubuntu-lab:~$ sudo cat /etc/environment
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
Class="CC-5A"
```

4. Re-login / open new shell — show Class and `PATH` together

```
hajra@ubuntu-lab:~$ echo $Class
CC-5A
hajra@ubuntu-lab:~$ echo "$PATH"
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
```

5. Create welcome script and make it executable

```
hajra@ubuntu-lab:~$ cat > ~/welcome <<'EOF'
> #!/bin/bash
> echo "Welcome to Cloud Computing $USER"
> EOF
hajra@ubuntu-lab:~$ chmod +x ~/welcome
```

6. Run the script using `./welcome`

```
hajra@ubuntu-lab:~$ cd ~
hajra@ubuntu-lab:~$ ./welcome
Welcome to Cloud Computing hajra
```

7. Add `PATH` entry to `~/.bashrc`

```
PATH=$PATH:~
-- INSERT --
```

8. Apply changes and run welcome without `./`

```
hajra@ubuntu-lab:~$ cd ~
hajra@ubuntu-lab:~$ welcome
Welcome to Cloud Computing hajra
```

Task 5 — Block and Allow SSH Using UFW (Firewall)

Goal:

Use `ufw` to deny and allow SSH access, then verify the connectivity changes from the Windows host.

1. Enable UFW and show its status

```
hajra@ubuntu-lab:~$ sudo ufw enable
[sudo] password for hajra:
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
hajra@ubuntu-lab:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip
```

2. Deny SSH (port 22) and show numbered status

```
hajra@ubuntu-lab:~$ sudo ufw deny 22/tcp
Rule added
Rule added (v6)
hajra@ubuntu-lab:~$ sudo ufw status numbered
Status: active

To Action From
--
[ 1] 22/tcp DENY IN Anywhere
[ 2] 22/tcp (v6) DENY IN Anywhere (v6)
```

3. From Windows host — attempt SSH (should fail)

```
hajra@ubuntu-lab:~$ exit
logout
Connection to 192.168.161.129 closed.
PS C:\Users\HP> ssh hajra@192.168.161.129
ssh: connect to host 192.168.161.129 port 22: Connection timed out
PS C:\Users\HP>
```

4. Allow SSH again, reload UFW, and show status

```
The list of available updates is more than a week old.
To check for new updates run: sudo apt update

hajra@ubuntu-lab:~$ sudo ufw allow 22/tcp
[sudo] password for hajra:
Rule updated
Rule updated (v6)
hajra@ubuntu-lab:~$ sudo ufw reload
firewall reloaded
hajra@ubuntu-lab:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
Anywhere on vxlan.calico ALLOW Anywhere
Anywhere on cali+ ALLOW Anywhere (v6)
22/tcp (v6) ALLOW Anywhere (v6)
Anywhere (v6) on vxlan.calico ALLOW Anywhere (v6)
Anywhere (v6) on cali+ ALLOW Anywhere (v6)

Anywhere ALLOW OUT Anywhere on vxlan.calico
Anywhere (v6) ALLOW OUT Anywhere (v6) on vxlan.calico
Anywhere (v6) ALLOW OUT Anywhere (v6) on cali+
```

5. From Windows host — attempt SSH again (should succeed)

```
PS C:\Users\HP> ssh hajra@192.168.161.129
Warning: Permanently added '192.168.161.129' (SSH-2.0-ubuntu_24.04.3 LTS (GNU/Linux 6.8.0-60-generic x86_64))
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-60-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat 15 Nov 14:01:30 UTC 2025

System load: 1.31 Processes: 318
Usage of /: 47.2K of 37.29GB Users logged in: 1
Memory usage: 5% IP address for ens33: 192.168.161.129
Swap usage: 11%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.
   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

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12 additional security updates can be applied with the Apps
   team more about enabling the Apps service at https://ubuntu.com/esm

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Sat Nov 15 08:51:04 2025 from 192.168.161.1
hajra@ubuntu-lab:~$
```

Task 6 — Configure SSH Key-Based Login from Windows Host

Goal:

Copy the Windows client's public key into the Ubuntu server's `~/.ssh/authorized_keys` to enable passwordless SSH.

A. Windows Host (Client-Side Steps)

1. Generate SSH key pair and list `.ssh` directory

```
Windows PowerShell
PS C:\Users\HP> ssh-keygen -t ed25519 -f $env:USERPROFILE\.ssh\id_lab7 -C "lab_key"
generating public/private ed25519 key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\HP\.ssh\id_lab7
Your public key has been saved in C:\Users\HP\.ssh\id_lab7.pub
The key fingerprint is:
SHA256:20H6S0Xun1x0ghyVdp+Hkkm0BTJ+c6llnmUthwFgo lab_key
The key's randomart image is:
+--[ED25519 256]--+
|  .oo. +. +      |
| .+o. +. X + o   |
| .++ . X + .     |
| .++ . X + .     |
| E.+oS O .       |
| .o+o.X +        |
| ooooo =         |
| . . .           |
| oo.             |
+---[SHA256]-----+

PS C:\Users\HP> ls $env:USERPROFILE\.ssh

Directory: C:\Users\HP\.ssh

Mode                LastWriteTime         Length Name
----                -
-a-----         10/3/2025 11:59 AM           419 id_ed25519
-a-----         10/3/2025 11:59 AM           110 id_ed25519.pub
-a-----        11/15/2025  7:26 PM           444 id_lab7
-a-----        11/15/2025  7:26 PM            90 id_lab7.pub
-a-----         10/3/2025 12:09 PM          1671 known_hosts
-a-----         10/3/2025 12:09 PM           938 known_hosts.old
```

2. Show the public key content

```
PS C:\Users\HP> type $env:USERPROFILE\.ssh\id_lab7.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIK13ZoDhdGEmrOlZCEFIyBfO3NL649vzb6VEmAL00rMk lab_key
```

3. Clear the known_hosts file and verify it is empty

```
PS C:\Users\HP> Clear-Content $env:USERPROFILE\.ssh\known_hosts
PS C:\Users\HP> type $env:USERPROFILE\.ssh\known_hosts
PS C:\Users\HP>
```

4. Connect to Ubuntu server — capture host-key acceptance

```
PS C:\Users\HP> ssh hajra@192.168.161.129
The authenticity of host '192.168.161.129 (192.168.161.129)' can't be established.
ED25519 key fingerprint is SHA256:rsuam67+jxhroK03bBPRkghldskqSusptQ25Up3w.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.161.129' (ED25519) to the list of known hosts.
hajra@192.168.161.129's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat 15 Nov 14:36:17 UTC 2025

System load:  0.76          Processes:      286
Usage of /:   47.2% of 37.29GB   Users logged in: 1
Memory usage: 50%           IPv4 address for ens33: 192.168.161.129
Swap usage:   13%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
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   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

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To check for new updates run: sudo apt update

Last login: Sat Nov 15 14:01:31 2025 from 192.168.161.1
hajra@ubuntu-lab: ~$
```

5. After login, show known_hosts file updated

```
PS C:\Users\HP> type $env:USERPROFILE\.ssh\known_hosts
192.168.161.129 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIK13ZoDhdGEmrOlZCEFIyBfO3NL649vzb6VEmAL00rMk lab_key
192.168.161.129 ssh-rsa AAAAB3QzaC1yc2EAAAADAQABAAQGDGKfL72KStt5Vhtms+rdf17VjYlSVhSAdRuxABANRAGQGWupUCLANos0RLW0Yf+avfjKcNdhjEIDVq2ZDepKAp123/KB48msIXCpVVR6y13CVX1j2KTRHYXC1ao3o7J
xwvdiYkHmIR/fKfr7thEyX1ORUCe2zFxcg2QYVQe0saBPd4dcrowlxfdtts13w/Bq505BCquoP7+n6ja/BGng0sf93yjkqB2X3JN3lIas0msF50w1BCr04ME10tg7zmtDfqtzuwWMC7xf/eG+xf5XGZDAMHh.eJcXln4AIPVtkk+v6jkt
0Y0PMPiPaepASarXWcfJlMfJKSwd7su/KHfL4fopfRLtZIEfmo0KND3B1X0u87pYR8Rjgkxt1ZYAS212gpl1IMQCHzoxutVUPax1VKPVRxraozK3A00rU6GLX1T98G11uB1B8jH0bc1cyNEHK/lpGp1p1IKnnC9ovGwd3vrmKPL1x5G1k
-
192.168.161.129 ed25519-sha2-nistp256 AAAAEZVjZmhlXkovT1t1mlzdh4yNTYAABBBKtMq1on63g8RHF0bS5Z+chx0dhf09aw1Sr1df7ps/1+P117Zg0K07NDp+ln28qatg3BLwdh7uWg4ED0XLE=
PS C:\Users\HP>
```

B. Ubuntu Server (Server-Side Steps)

6. Prepare .ssh directory and clear authorized_keys

```
hajra@ubuntu-lab:~$ mkdir -p ~/.ssh
hajra@ubuntu-lab:~$ chmod 700 ~/.ssh
hajra@ubuntu-lab:~$ > ~/.ssh/authorized_keys
hajra@ubuntu-lab:~$
```

7. Add the public key, set permissions, and display file

```
hajra@ubuntu-lab:~$ echo "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIK13ZoDhdGEmrOlZCEFIyBfO3NL649vzb6VEmAL00rMk lab_key" >> ~/.ssh/authorized_keys
hajra@ubuntu-lab:~$ chmod 600 ~/.ssh/authorized_keys
hajra@ubuntu-lab:~$ cat ~/.ssh/authorized_keys

ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIK13ZoDhdGEmrOlZCEFIyBfO3NL649vzb6VEmAL00rMk lab_key
```

C. Test Passwordless SSH

8. From Windows — confirm passwordless login

```
PS C:\Users\HP> ssh hajra@192.168.161.129
hajra@192.168.161.129's password: ■
```

9. Also test using explicit identity file

```
PS C:\Users\HP> ssh -i $env:USERPROFILE\.ssh\id_lab7 hajra@192.168.161.129
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat 15 Nov 15:31:13 UTC 2025

System load:  0.72          Processes:      290
Usage of /:   47.2% of 37.29GB   Users logged in: 1
Memory usage: 61%           IPv4 address for ens33: 192.168.161.129
Swap usage:   12%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.
   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

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To check for new updates run: sudo apt update

Last login: Sat Nov 15 15:25:21 2025 from 192.168.161.1
hajra@ubuntu-lab:~$
```

Exam Evaluation Questions

Q1 — Quick Environment Audit

Objective:

Demonstrate you can inspect the current environment and extract a few key variables.

Actions & Evidence:

1. Display all environment variables:

```
hajra@ubuntu-lab:~$ printenv
SHELL=/bin/bash
PWD=/home/hajra
LOGNAME=hajra
XDG_SESSION_TYPE=tty
HOME=/home/hajra
LANG=en_GB.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:bd=40;33;01:cd=40;33;01:or=01;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lzh=01;31:*.lzm=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lzo=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzt=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.t2=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;31:*.jpg=01;31:*.jpeg=01;31:*.mjpg=01;31:*.mjpeg=01;31:*.gif=01;31:*.bmp=01;31:*.pbm=01;31:*.pgm=01;31:*.ppm=01;31:*.tga=01;31:*.xbm=01;31:*.xpm=01;31:*.tif=01;31:*.tiff=01;31:*.png=01;31:*.svg=01;31:*.svgz=01;31:*.mng=01;31:*.pcc=01;31:*.mov=01;31:*.mpg=01;31:*.mpeg=01;31:*.m2v=01;31:*.mkv=01;31:*.webm=01;31:*.webp=01;31:*.ogm=01;31:*.mp4=01;31:*.m4v=01;31:*.mp4v=01;31:*.vob=01;31:*.qt=01;31:*.nuv=01;31:*.wmv=01;31:*.asf=01;31:*.rm=01;31:*.rmvb=01;31:*.flc=01;31:*.avi=01;31:*.flv=01;31:*.gl=01;31:*.dl=01;31:*.xcf=01;31:*.xwd=01;31:*.yuv=01;31:*.cgm=01;31:*.emf=01;31:*.ogv=01;31:*.ogx=01;31:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.000=00;90:*.bak=00;90:*.crdownload=00;90:*.dpg=00;90:*.dpg-new=00;90:*.dpg-old=00;90:*.dpg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.rpm-new=00;90:*.rpmorig=00;90:*.rpmsave=00;90:*.swp=00;90:*.tmp=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
SSH_CONNECTION=192.168.161.1 50467 192.168.161.129 22
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=hajra
SHLVL=1
XDG_SESSION_ID=3
XDG_RUNTIME_DIR=/run/user/1000
SSH_CLIENT=192.168.161.1 50467 22
XDG_DATA_DIRS=/usr/share/gnome:/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
SSH_TTY=/dev/pts/0
/usr/bin/printenv
```

2. Filter for PATH, LANG, and PWD (run 3 commands, capture together):

```
hajra@ubuntu-lab:~$ printenv | grep PATH
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/home/hajra
hajra@ubuntu-lab:~$ printenv | grep LANG
LANG=en_GB.UTF-8
hajra@ubuntu-lab:~$ printenv | grep PWD
PWD=/home/hajra
```

Q2 — Short-Lived Student Info

Objective:

Show how temporary environment variables behave (session-scoped).

Actions & Evidence:

1. Set three variables consecutively and capture commands + execution:

```
hajra@ubuntu-lab:~$ export STUDENT_NAME="John Doe"
hajra@ubuntu-lab:~$ export STUDENT_ROLL_NUMBER="12345"
hajra@ubuntu-lab:~$ export STUDENT_SEMESTER="3"
```

2. Print all three variables (grouped) and capture output:

```
hajra@ubuntu-lab:~$ echo "$STUDENT_NAME"
John Doe
hajra@ubuntu-lab:~$ echo "$STUDENT_ROLL_NUMBER"
12345
hajra@ubuntu-lab:~$ echo "$STUDENT_SEMESTER"
3
```

3. Use `printenv | grep` to show any STUDENT variables:

```
hajra@ubuntu-lab:~$ printenv | grep '^STUDENT_'
STUDENT_NAME=John Doe
STUDENT_SEMESTER=3
STUDENT_ROLL_NUMBER=12345
```

4. Exit shell, open new terminal, show variables are gone:

```
hajra@ubuntu-lab:~$ echo "$STUDENT_NAME"
John Doe
hajra@ubuntu-lab:~$ printenv | grep '^STUDENT_'
STUDENT_NAME=John Doe
STUDENT_SEMESTER=3
STUDENT_ROLL_NUMBER=12345
```


Q3 — Make It Sticky (Persistent Student Info)

Objective:

Demonstrate persistence of environment variables across sessions via `~/.bashrc`.

Actions & Evidence:

1. Edit `~/.bashrc` and append the three `STUDENT_*` exports. Capture **editor screenshot**:

```
export STUDENT_NAME="John Doe"
export STUDENT_ROLL_NUMBER="12345"
export STUDENT_SEMESTER="3"

-- INSERT --
```

2. Reload shell config and verify variables + `printenv` grep together:

```
hajra@ubuntu-lab:~$ source ~/.bashrc
ho "$STUDENT_ROLL_NUMBER"
echo "$STUDENT_SEMESTER"
printenv | grep '^STUDENT_'
hajra@ubuntu-lab:~$ echo "$STUDENT_NAME"
John Doe
hajra@ubuntu-lab:~$ echo "$STUDENT_ROLL_NUMBER"
12345
hajra@ubuntu-lab:~$ echo "$STUDENT_SEMESTER"
3
hajra@ubuntu-lab:~$ printenv | grep '^STUDENT_'
STUDENT_NAME=John Doe
STUDENT_SEMESTER=3
STUDENT_ROLL_NUMBER=12345
hajra@ubuntu-lab:~$
```

3. Close and reopen terminal — verify `STUDENT NAME` persists:

```
hajra@ubuntu-lab:~$ echo "$STUDENT_NAME"
John Doe
hajra@ubuntu-lab:~$ printenv | grep '^STUDENT_'
STUDENT_NAME=John Doe
STUDENT_SEMESTER=3
STUDENT_ROLL_NUMBER=12345
hajra@ubuntu-lab:~$
```

Q4 — Firewall Rules: Block and Restore Ping (ICMP)

Objective:

Block ping (ICMP echo) traffic using `ufw` and then re-allow it; show effect from a client.

Actions & Evidence:

1. Enable `ufw` and show status (group commands):

```
hajra@ubuntu-lab:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
hajra@ubuntu-lab:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
Anywhere on vxlan.calico ALLOW Anywhere
Anywhere on cali+ ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
Anywhere (v6) on vxlan.calico ALLOW Anywhere (v6)
Anywhere (v6) on cali+ ALLOW Anywhere (v6)

Anywhere ALLOW OUT Anywhere on vxlan.calico
Anywhere ALLOW OUT Anywhere on cali+
Anywhere (v6) ALLOW OUT Anywhere (v6) on vxlan.calico
Anywhere (v6) ALLOW OUT Anywhere (v6) on cali+
```

2. Block ping (ICMP) and show numbered status:

```
hajra@ubuntu-lab:~$ sudo iptables -A INPUT -p icmp --icmp-type echo-request -j DROP
hajra@ubuntu-lab:~$ sudo ufw status numbered
Status: active

To Action From
--
[ 1] 22/tcp ALLOW IN Anywhere
[ 2] Anywhere on vxlan.calico ALLOW IN Anywhere
[ 3] Anywhere ALLOW OUT Anywhere on vxlan.calico (out)
[ 4] Anywhere on cali+ ALLOW IN Anywhere
[ 5] Anywhere ALLOW OUT Anywhere on cali+ (out)
[ 6] 22/tcp (v6) ALLOW IN Anywhere (v6)
[ 7] Anywhere (v6) on vxlan.calico ALLOW IN Anywhere (v6)
[ 8] Anywhere (v6) ALLOW OUT Anywhere (v6) on vxlan.calico (out)
[ 9] Anywhere (v6) on cali+ ALLOW IN Anywhere (v6)
[10] Anywhere (v6) ALLOW OUT Anywhere (v6) on cali+ (out)
```

3. From Windows host (or another client) — attempt ping while blocked:

```
PS C:\Users\HP> ping 192.168.161.129

Pinging 192.168.161.129 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.161.129:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PS C:\Users\HP> _
```

4. Re-allow ping and show allow/reload/status together:

```
hajra@ubuntu-lab:~$ sudo iptables -A INPUT -p icmp --icmp-type echo-request -j ACCEPT
[sudo] password for hajra:
hajra@ubuntu-lab:~$ _
```

5. From client, ping server again — capture successful replies:

```
PS C:\Users\HP> ping 192.168.161.129

Pinging 192.168.161.129 with 32 bytes of data:
Reply from 192.168.161.129: bytes=32 time<1ms TTL=64
Reply from 192.168.161.129: bytes=32 time<1ms TTL=64
Reply from 192.168.161.129: bytes=32 time=1ms TTL=64
Reply from 192.168.161.129: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.161.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Users\HP>
```
